Problem solving with sequences

1 Which would describe the sequence 1,3,6,10,15, …? Tick **1** correct answer

- [ ] Arithmetic
- [ ] Linear
- [x] Non-linear

2 The 2^{nd} term of an arithmetic sequence is 11 and the 6^{th} term is 47. What is the 1^{st} term of the sequence? Tick **1** correct answer

- [ ] 0
- [ ] 1
- [x] 2
- [ ] 3
- [ ] 4

3 Buses leave Wallsend for Byker at regular intervals. The second bus of the day leaves at 6:44 am. You just missed the seventh bus of the day which left at 8:04 am. What time is the next bus? Tick **1** correct answer

- [ ] 8:14 am
- [x] 8:20 am
- [ ] 8:24 am
- [ ] 8:30 am
- [ ] 8:34 am

4 The 24^{th} term of an arithmetic sequence is 281. The 37^{th} term is 437. Which calculation will be the starting point to finding the n^{th} term of the sequence? Tick **1** correct answer

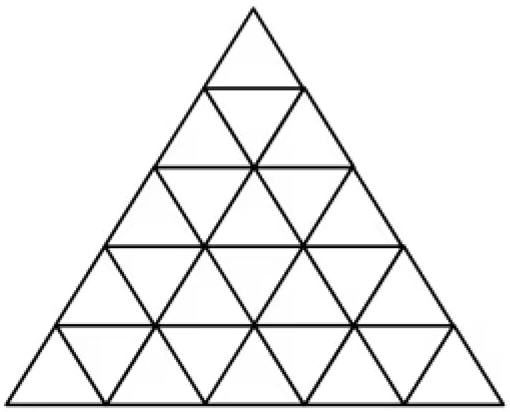
- [x] $\frac{437 - 281}{37 - 24}$
- [ ] $\frac{437 - 281}{24 - 37}$
- [ ] $\frac{281 - 437}{37 - 24}$
- [ ] $\frac{37 - 24}{437 - 281}$
- [ ] $\frac{24 - 37}{437 - 281}$

5 What is the n^{th} term of this sequence? Tick **1** correct answer

$$\frac{2}{3}, \frac{9}{8}, \frac{16}{13}, \frac{23}{18}, \ldots$$

- [ ] $\frac{n + 7}{n + 5}$
- [ ] $\frac{7n + 2}{5n + 3}$
- [ ] $\frac{7n + 5}{5n + 2}$
- [x] $\frac{7n - 5}{5n - 2}$
- [ ] It's not an arithmetic (linear) sequence we can't write an n^{th} term.

6 How many triangles are there in this pattern? Don't forget to count all triangles of all sizes.
Tick **1** correct answer



- [ ] 50
- [x] 48
- [ ] 45
- [ ] 38
- [ ] 35